

Lowering of vapour pressure

21. When a substance is dissolved in a solvent, the vapour pressure of the solvent is decreased. This results in
- (a) An increase in the boiling point of the solution
 - (b) A decrease in the boiling point of solvent
 - (c) The solution having a higher freezing point than the solvent
 - (d) The solution having a lower osmotic pressure than the solvent
22. The vapour pressure of a liquid depends on
- (a) Temperature but not on volume
 - (b) Volume but not on temperature
 - (c) Temperature and volume
 - (d) Neither on temperature nor on volume
23. Which one of the statements given below concerning properties of solutions, describes a colligative effect
- (a) Boiling point of pure water decreases by the addition of ethanol
 - (b) Vapour pressure of pure water decreases by the addition of nitric acid
 - (c) Vapour pressure of pure benzene decreases by the addition of naphthalene
 - (d) Boiling point of pure benzene increases by the addition of toluene
24. The atmospheric pressure is sum of the
- (a) Pressure of the biomolecules
 - (b) Vapour pressure of atmospheric constituents
 - (c) Vapour pressure of chemicals and vapour pressure of volatiles
 - (d) Pressure created on to atmospheric molecules
25. The vapour pressure of pure liquid A is 0.80 atm . On mixing a non-volatile B to A, its vapour pressure becomes 0.6 atm . The mole fraction of B in the solution is
- (a) 0.150
 - (b) 0.25
 - (c) 0.50
 - (d) 0.75
26. Lowering of vapour pressure is highest for
- (a) Urea
 - (b) $0.1M$ glucose
 - (c) $0.1MMgSO_4$
 - (d) $0.1MBaCl_2$



27. An aqueous solution of glucose was prepared by dissolving 18 g of glucose in 90 g of water. The relative lowering in vapour pressure is
(a) 0.02 (b) 1
(c) 20 (d) 180
28. "Relative lowering in vapour pressure of solution containing non-volatile solute is directly proportional to mole fraction of solute". Above statement is
(a) Henry law
(b) Dulong and Petit law
(c) Raoult's law
(d) Le-Chatelier's principle
29. An ideal solution was obtained by mixing methanol and ethanol. If the partial vapour pressure of methanol and ethanol are 2.619 kPa and 4.556 kPa respectively, the composition of the vapour (in terms of mole fraction) will be
(a) 0.635 methanol, 0.365 ethanol
(b) 0.365 methanol, 0.635 ethanol
(c) 0.574 methanol, 0.326 ethanol
(d) 0.173 methanol, 0.827 ethanol
30. The vapour pressure of two liquids P and Q are 80 and 600 torr, respectively. The total vapour pressure of solution obtained by mixing 3 mole of P and 2 mole of Q would be
(a) 140 torr (b) 20 torr
(c) 68 torr (d) 72 torr
31. The vapour pressure of benzene at a certain temperature is 640 mm of Hg. A non-volatile and non-electrolyte solid weighing 2.175 g is added to 39.08 g of benzene. The vapour pressure of the solution is 600 mm of Hg. What is the molecular weight of solid substance
(a) 49.50 (b) 59.6
(c) 69.5 (d) 79.8
32. Which one of the following is the expression of Raoult's law
(a) $\frac{p-p_s}{p} = \frac{n}{n+N}$ (b) $\frac{p_s-p}{p} = \frac{N}{N+n}$
(c) $\frac{p-p_s}{p_s} = \frac{N}{N-n}$ (d) $\frac{p_s-p}{p_s} = \frac{N-n}{N}$
 p = vapour pressure of pure solvent
 p_s = vapour pressure of the solution
 n = number of moles of the solute
 N = number of moles of the solvent
33. Which has maximum vapour pressure
(a) HI (b) HBr
(c) HCl (d) HF
34. When a non-volatile solute is dissolved in a solvent, the relative



- lowering of vapour pressure is equal to
- (a) Mole fraction of solute
(b) Mole fraction of solvent
(c) Concentration of the solute in grams per litre
(d) Concentration of the solute in grams 100 ml
35. 60 gm of Urea (Mol. wt 60) was dissolved in 9.9 moles, of water. If the vapour pressure of pure water is P_o , the vapour pressure of solution is
(a) $0.10 P_o$ (b) $1.10 P_o$
(c) $0.90 P_o$ (d) $0.99 P_o$
36. The vapour pressure of water at $20^\circ C$ is 17.54 mm. When 20g of a non-ionic, substance is dissolved in 100g of water, the vapour pressure is lowered by 0.30 mm. What is the molecular weight of the substances
(a) 210.2 (b) 206.88
(c) 215.2 (d) 200.8
37. In an experiment, 1 g of a non-volatile solute was dissolved in 100 g of acetone (mol. mass = 58) at 298K. The vapour pressure of the solution was found to be 192.5 mm Hg. The molecular weight of the solute is (vapour pressure of acetone = 195 mm Hg)
- (a) 25.24 (b) 35.24
(c) 45.24 (d) 55.24
38. How many grams of CH_3OH should be added to water to prepare 150ml solution of 2M CH_3OH
(a) 9.6 (b) 2.4
(c) 9.6×10^3 (d) 2.4×10^3
39. The vapour pressure of a solvent decreased by 10mm of mercury, when a non-volatile solute was added to the solvent. The mole fraction of the solute in the solution is 0.2. What should be the mole fraction of the solvent, if decrease in the vapour pressure is to be 20mm of mercury
(a) 0.8 (b) 0.6
(c) 0.4 (d) 0.2
40. For a dilute solution, Raoult's law states that
(a) The lowering of vapour pressure is equal to mole fraction of solute
(b) The relative lowering of vapour pressure is equal to mole fraction of solute
(c) The relative lowering of vapour pressure is proportional to the amount of solute in solution



- (d) The vapour pressure of the solution is equal to the mole fraction of solvent
41. The vapour pressure of a solvent A is 0.80 atm . When a non-volatile substance B is added to this solvent its vapour pressure drops to 0.6 atm . What is mole fraction of B in solution
- (a) 0.25
(b) 0.50
(c) 0.75
(d) 0.90
42. Determination of correct molecular mass from Raoult's law is applicable to
- (a) An electrolyte in solution
(b) A non-electrolyte in a dilute solution
(c) A non-electrolyte in a concentrated solution
(d) An electrolyte in a liquid solvent
43. If two substances A and B have $P_A^0 : P_B^0 = 1 : 2$ and have mole fraction in solution $1 : 2$ then mole fraction of A in vapours
- (a) 0.33
(b) 0.25
(c) 0.52
(d) 0.2
44. A dry air is passed through the solution, containing the 10 gm of solute and 90 gm of water and then it pass through pure water. There is the depression in weight of solution wt by 2.5 gm and in weight of pure solvent by 0.05 gm . Calculate the molecular weight of solute
- (a) 50
(b) 180
(c) 100
(d) 25

